ELECTRICAL MACHINES · CH1–2

Power, phasors and complex power

Cheat sheet · Day 01 · Sat 24 Oct · pack M1

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
M1  POWER, PHASORS, COMPLEX POWER  (rms throughout; V1 = V1/_0 is the reference)
phi = angle of I relative to V.  lag (inductive): phi<0, Q>0.  lead (capacitive): phi
w = 2*pi*n/60 [rad/s]   (rpm ~ 10 x rad/s)    P_M = τ*w    τ = F*r, F = m*g
eta = Pout/Pin   Ploss = Pin - Pout = Pin(1-eta)   motor: Pin electrical, Pout = τ*w
     motor rating = OUTPUT (shaft) power; 3ph nameplate = line V, line I, total P; as
DC source:  E = Voc,  Rs = (E - V)/I,  V = E - I*Rs,  Pload = V*I,  Pmax at RL = Rs
1ph: P = V1 I1 cos(phi)   Q = V1 I1 sin(-phi)   S = V1 I1     Vpk = sqrt2 * Vrms
3ph: S = 3 V1 I1 = sqrt3 VL IL,  P = sqrt3 VL IL cos(phi),  pf(3ph) = pf(1 phase)
p(t) = P + S cos(2wt + phi):  average P, freq 2f, min = P - S (negative if Q != 0)
|phi| = (dt / T) * 360 deg; current peaks earlier => leading
Z:  R,  jwL,  -j/(wC).   Series: add Z.  Parallel: Z1Z2/(Z1+Z2) or add branch current
S = V1 * conj(I1) = P + jQ = S/_(-phi)    pf = P/S = cos(phi)  (STATE lead/lag)
Q = +/- sqrt(S^2 - P^2): sign from physics (L +, C -)
Sum P's and sum Q's over loads; never sum S's.
SERIES RL (same I):   V1^2 = VR^2 + VL^2,  R = P/I^2,  X = Q/I^2,  P:Q = R:X
PARALLEL RL (same V): I1^2 = IR^2 + IL^2,  R = V^2/P,  X = V^2/Q,  P:Q = X:R
L = X/w    C = 1/(w|X|)    series RLC: |VL| or |VC| may exceed V1; f0 = 1/(2 pi sqrt(
Check every answer:  P = sum I^2 R  and  Q = sum I^2 X (signed)
Calculator: if Re < 0, add/subtract 180 deg to the atan result
PF correction (see M2): a parallel C makes Q negative; P is unchanged; supply S and I
```